

Perceptual ambiguity of the self – dynamics connect brain and behaviour

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ARTICLE INFO

Keywords:

Self-other discrimination
EEG
Perceptual ambiguity
Decision-making
Drift diffusion modelling
Neural dynamics
Neurobehavioural mechanisms

ABSTRACT

When the distinction between self and other is blurred, the self-other boundary is weakened, introducing perceptual ambiguity. The neural and behavioural mechanisms of resolving this ambiguity, and their interaction, are not known. To address this, we used a facial morphing task recorded in EEG, where participants' faces were morphed with the face of another in different proportions. Ambiguous images elicited different neural and behavioural dynamics than unambiguous images. Firstly, drift diffusion modelling of response behaviour revealed slower drift rates/evidence accumulation for ambiguous vs. unambiguous stimuli, which was accompanied by lower task-state neural measures of peak frequency sliding (PFS) in the theta band. Secondly, entropy of response times (RTs) and spectral entropy (SE) were higher for ambiguous vs. unambiguous faces, implying greater unpredictability and increased processing demand in both brain and behaviour. Notably, we identified neurobehavioural relationships for both dynamic measures. Resting-state theta PFS and ambiguous drift rates were significantly related, suggesting that stronger and faster internal information uptake during rest was related to better evidence accumulation for ambiguous faces. Moreover, SE on the neural level was related to behavioural entropy: smaller SE increase from rest to task was linked to lower RT entropy in the ambiguous condition, and thus better allocation of cognitive resources. Together, we demonstrate speed of information uptake and entropy/processing capacity are key in efficiently resolving self-other ambiguity, serving as the "common currency" (Northoff et al., 2020, 2025) of neural and behavioural levels of processing.

1. Introduction

Every time people glance at their reflection, there is hardly any doubt that the face staring back is their own. Mirror self-recognition is an automatic process that defines self-identity (Volpara et al., 2022). But how stable is our ability to discriminate what is self and what is not when our facial image is confounded with that of another? When some of the features staring back in the mirror are not our own? In the present study, we investigated perceptual ambiguity in self-face images to examine the dynamic neural and behavioural processes involved in dealing with ambiguous decision-making. We used self-other face morphs as a high-salience stimulus for this analysis.

1.1. The ambiguity problem in decision-making

All decisions are plagued by various levels of uncertainty. People are intrinsically averse to ambiguity, a concept which has guided decision-making and economic theory (Ellsberg, 1961; Huettel et al., 2006). In

ambiguous decisions, the probability of outcomes is unknown, which elicits a strong aversion in comparison to decisions in which the risk is known (Ellsberg, 1961; Wu et al., 2021). Overall, high ambiguity is generally harder to interpret and process and can even elicit negative emotional responses (Jakesch et al., 2013). The greater the ambiguity between alternatives, the more difficult the decision becomes, requiring greater neural (Maksimenko et al., 2019; Maksimenko et al., 2020; Shen et al., 2019) and computational resources (Sun et al., 2025) to evaluate outcomes. In other words, ambiguity is expensive, in terms of both cognitive resources and time.

How can we explain the increased cost of processing ambiguity? An ambiguous stimulus has at least two possible interpretations. The perceptual representations of each alternative are thought to be in constant flux in the perceptual system, with each one coded by a different population of neurons (Klink et al., 2012). Therefore, the ambiguity of the stimulus is resolved by the interpretation with the strongest neuronal representation, which can be affected by any number of top-down and bottom-up factors (Klink et al., 2012). As a result of the

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<https://doi.org/10.1016/j.neuroimage.2026.121839>

Received 11 November 2025; Received in revised form 28 February 2026; Accepted 2 March 2026

Available online 3 March 2026

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increased computational cost involved in evaluating multiple decision outcomes (Klink et al., 2012; Esposito et al., 2023; Karimi-Rouzbahani et al., 2021), response times (RTs) to ambiguous stimuli are longer than for unambiguous stimuli (Maksimenko et al., 2019). Neurally, the increased allocation of cognitive and perceptual resources for processing high ambiguity is reflected by higher theta power in frontal areas (Maksimenko et al., 2019) and is encoded in the strength of a late positive potential (LPP) elicited during decision-making (Sun et al., 2017). In sum, there is strong evidence that ambiguous stimuli are handled differently in decision-making than unambiguous stimuli.

A situation in which ambiguity is often overlooked is the differentiation between internal processes, or the self, and external processes originating from others or the environment (de Bézenac et al., 2018). This ability to discriminate “self” and “other” is a key developmental milestone, emerging in childhood as early as 15 months (Volpara et al., 2022; Lewis and Ramsay, 2004). Self-face becomes an overlearned, familiar stimulus, its mental representation formed based on a reflection of our true identity (Volpara et al., 2022). It is through this self-representation that we identify familiar and unknown faces and are able to maintain healthy self-perception. de Bézenac et al. (de Bézenac

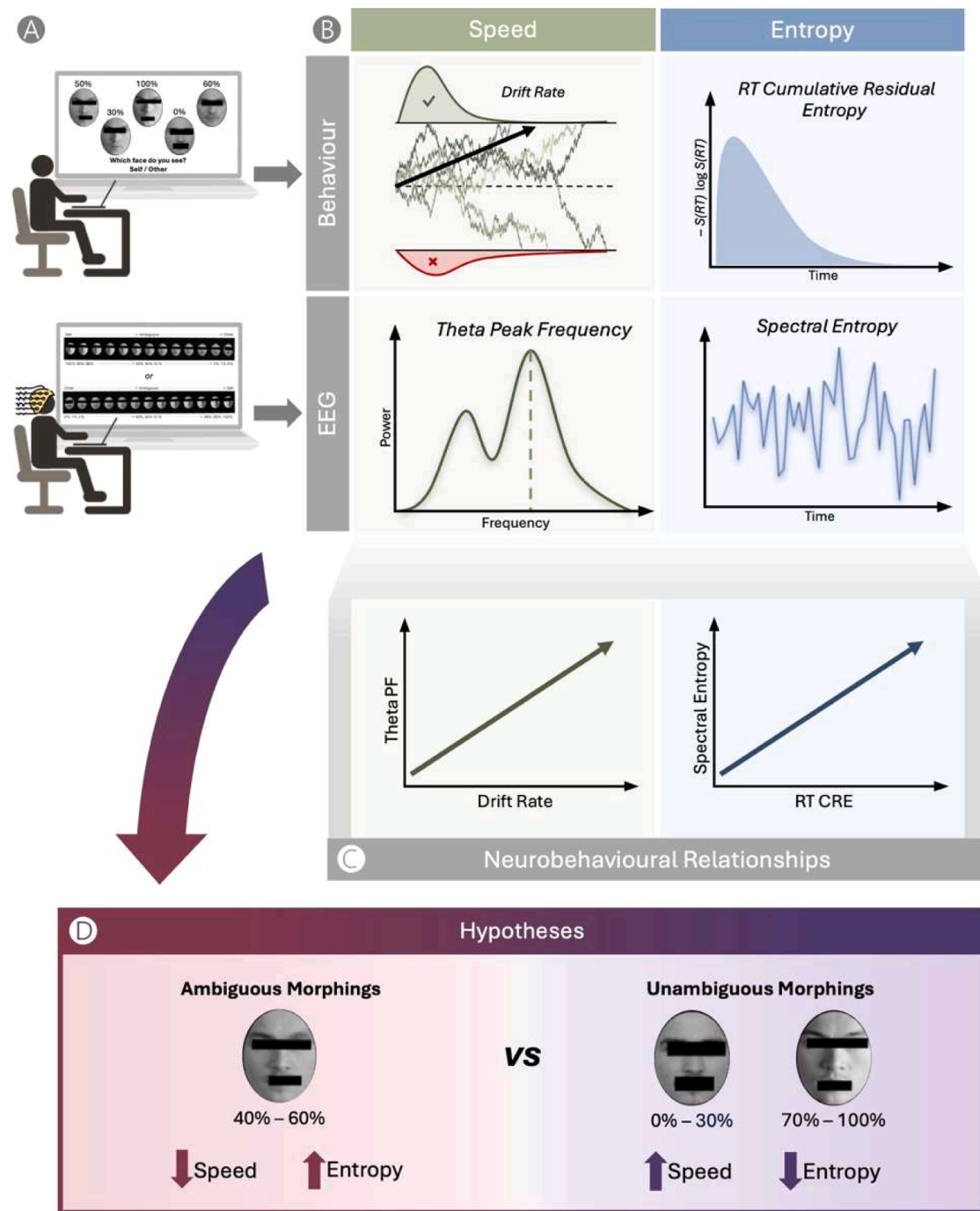


Fig. 1. Conceptual Framework. Schematic of task and analysis design. A) Illustration of the discontinuous and continuous tasks, highlighting the difference in presentation of the morphed images between the two. B) Measures of interest indexing speed (green) and entropy (blue) for each task with analysis type in grey. C) Planned linear modelling analyses to link neurodynamic and behavioural measures. D) All measures are computed separately for ambiguous and unambiguous morphings, for the given morphing steps in terms of % self. Predictions for the contrasts between conditions are shown along the bottom of D).

et al., 2018) argued for the importance of making decisions under self-other ambiguity, stating that being exposed to this ambiguity allows individuals to be more attuned to the self-other boundary. Taking advantage of such an overlearned, automatically processed stimulus, we seek to identify the neural and behavioural mechanisms behind this perceptual ambiguity, proposed to be essential to the development of critical thinking and agency (de Bézenac et al., 2018).

1.2. Neural and behavioural dynamics of ambiguous decision-making

Reiterating our initial question: how is self-recognition impacted when the boundaries between self and other are deliberately blurred? Using a facial morphing task where participants' faces are morphed with a stranger's in incremental percentages, we can shed light on this subject (Keenan et al., 2000; Kircher et al., 2001). The morphed images create a continuum, which, when presented continuously, have been used to study the emergence and persistence of self-face recognition (Tsakiris, 2008; Wolman et al., 2023). Similarly, by presenting the morphed images separately, the speed and accuracy of self-recognition at different morphing percentages can be assessed. In their 2016 paper, Bortolon et al. (Bortolon et al., 2016) showed that responses are faster and more accurate for morphing percentages that are unambiguous, i.e. stimuli where self-face is above 60 % or below 40 %. These results have been found in healthy controls as well as for people with schizophrenia (Bortolon et al., 2016), anorexia nervosa (Hiroto et al., 2016), and autism spectrum disorder (Wang et al., 2022).

Ultimately, discriminating between self and non-self can be computationally intensive, but, after all, it is simply a decision between two alternatives. The drift diffusion model (DDM) has been used extensively to model the cognitive and computational processes involved in two-alternative decision-making (Myers et al., 2022; Ratcliff, 1978; Ratcliff et al., 2016; Ratcliff and McKoon, 2008). The drift diffusion process consists of a single noisy accumulator that gathers evidence over time towards upper and lower boundaries representing the two choice outcomes, terminating at one of the two boundaries when a decision has been made (Fig. 1b). This diffusion process reflects and is based on RT and accuracy rates in a simple two-choice decision task (Ratcliff et al., 2016; Wiecki et al., 2013). Increasing the decision difficulty, or introducing ambiguity, slows evidence accumulation (drift rate) by decreasing the quality of evidence collected from the stimulus (Ratcliff and McKoon, 2008). This is supported by a new paper by Sun et al. (Sun et al., 2025) investigating ambiguity in facial expressions. They demonstrate over multiple experiments that there is a consistent negative relationship between ambiguity and drift rate, such that as ambiguity in facial expressions increased, drift rates decreased.

Intuitively, the dynamics of decision-making modelled by DDM may be related to the intrinsic dynamics of neural activity. The brain's internal resting state and stimulus-evoked activity can be described by spatiotemporal dynamics of the different frequency bands (alpha, theta, beta, etc.) of the neural signal (Northoff et al., 2024; Northoff et al., 2025). A commonly used descriptor is the frequency with the strongest power within a frequency band, the peak frequency (PF) (TH Grandy et al., 2013; Mierau et al., 2017). However, given the dynamic nature of neural activity, neural oscillations are not stationary, and frequency can fluctuate over time in response to changing internal or external variables (Cohen, 2014). Peak frequency sliding (PFS) is a method developed by Cohen (Cohen, 2014) that quantifies time-varying frequency dynamics as a phase-derived instantaneous frequency estimate at each time point. PFS is based on the fundamental property that changes in oscillatory frequency reflect modulation of underlying neural input and excitability, rather than changes in signal magnitude, and increases with increased input to a neural network (Cohen, 2014). This has been shown in both human EEG data as well as artificial neural networks (Cohen, 2014). Based on his previous work, Cohen (Cohen, 2014) proposed a hypothesis that the changes in frequency captured by PFS act as a gain-control mechanism. Lower PFS (slower oscillations) may increase

excitability to relatively weaker input signals allowing for faster, but noisier, spike patterns, while higher PFS (faster oscillations) may increase firing thresholds enabling more accurate and cautious responding.

Consequently, PFS has been used as a measure for neural coding of input intensity and information processing in spontaneous (resting-state) and stimulus-driven neural activity (Cohen, 2014; Hua et al., 2022; Wolff et al., 2019). PFS in the alpha band has been shown to track attention (Benwell et al., 2019) and the strength of external inputs, increasing in response to stimuli (Mierau et al., 2017; Cohen, 2014). As such, individual differences in alpha peak frequency have been robustly linked to working-memory capacity and general cognitive performance (e.g. (Klimesch et al., 1993; TH Grandy et al., 2013; Haegens et al., 2014; Rathee et al., 2020)), and are often interpreted as more stable, trait-like markers of cognitive efficiency. In contrast, research on PFS in the theta band has not been as widely explored (Hua et al., 2022; Wolff et al., 2019; Wolinski et al., 2018). Theta oscillations have been linked to internally-oriented cognition, involved in processes such as working memory (WM) (Wolinski et al., 2018; Moran et al., 2010) and cognitive control (Koshiyama et al., 2021; Rodriguez-Larios and Alaerts, 2019; Senoussi et al., 2022). Theta activity also plays an important role in self-face recognition; Kotlewska et al. (Kotlewska et al., 2023) and Miyakoshi et al. (Miyakoshi et al., 2010) demonstrated that there is a suppression of theta activity for self-face versus other facial stimuli. As a result, rather than tracking external inputs, theta PFS tracks the strength of internal inputs (Hua et al., 2022; Wolff et al., 2019). There is indeed evidence of a poststimulus increase in theta PFS for off-track thoughts (i.e. mind-wandering) reflecting a response to one's internal musings (Hua et al., 2022). Therefore, theta band oscillatory activity is a promising candidate for indexing the strength and speed of internally-driven information uptake for resolving perceptual ambiguity in self-other discrimination. Such decisions require the integration of stored facial representations under uncertainty (Klink et al., 2012) rather than the processing of stable external inputs typically attributed to alpha activity (Mierau et al., 2017; Cohen, 2014), making theta dynamics particularly relevant to this context. For these reasons, alpha PFS was not expected to capture the dynamic, internally-oriented ambiguity processing targeted here.

A complementary neurodynamic measure to PFS is the spectral entropy (SE) of a neural signal. Entropy measures the irregularity and uniformity of a signal, specifically the distribution of power across the signal's frequencies (Keshmiri, 2020). SE applies the principles of Shannon's entropy from information theory to assess the degree of complexity in the power spectrum of one's neural activity (Faust and Bairy, 2012; Shannon, 1948). A disordered neural signal will have higher values of SE, implying a greater number of possible neural states, while lower SE is characteristic of a distribution with less irregularity and greater certainty for a fewer number of potential activity patterns (Keshmiri, 2020; Shannon, 1948). Thus, SE has been used to index the information processing capacity of the system, defined by irregularity in its activity patterns. In the context of task-related activity, SE typically decreases in response to a stimulus, modulated from prestimulus (or resting-state) to poststimulus (Bachiller et al., 2014). Impaired SE modulation has been shown in clinical populations (Fernández-Linsenbarth et al., 2024; Hoshi et al., 2024; Iglesias-Tejedor et al., 2022; Molina et al., 2020), which suggests a deficit in neural synchronization that may contribute to cognitive symptoms and was even found to impact behavioural task performance (Molina et al., 2020). This research stresses the importance of proper SE modulation for stimulus processing and healthy brain function.

1.3. Aims and hypotheses

The current literature firmly establishes ambiguity's negative effects on decision-making and cognitive processing (Maksimenko et al., 2019; Maksimenko et al., 2020; Sun et al., 2017). However, despite the

importance of differentiating self vs. other (de Bézenac et al., 2018), none have explicitly investigated how ambiguity impacts this decision on both behavioural and neural grounds. Moreover, the relationship of neurodynamic measures such as PFS and SE to the dynamic drift diffusion process in decision-making has yet to be explored. In this paper, we bridge these two gaps by conducting a fine-grained analysis of neural and behavioural decision-making dynamics using DDM and spatiotemporal EEG analyses during a facial morphing task (Fig. 1a). The morphing task allows us to assess the neural and behavioural effects of perceptual ambiguity using self-face recognition.

Firstly, we analyze the impact of ambiguous faces on the speed of information uptake in decision-making. We expect slower information uptake for ambiguous vs. unambiguous stimuli due to the conflicting perceptual information within the images (Fig. 1d). We measure this behaviourally using DDM (Fig. 1b) to compare ambiguous and unambiguous drift rates, where we hypothesize slower evidence accumulation for ambiguous vs. unambiguous images. Neuronally, using theta peak frequency sliding (PFS) we measure the speed of internal input processing involved in disambiguating perceptual stimuli and self-face recognition (Fig. 1b). For ambiguous morphings, we predict lower theta PFS than for unambiguous morphings reflecting noisier neural activity and slower internal input processing in response to the stimuli.

Secondly, we use spectral entropy (SE) to measure the capacity of information processing in response to different levels of ambiguity (Keshmiri, 2020; Tian et al., 2017). Thus, we anticipate greater SE values for ambiguous stimuli in comparison to unambiguous stimuli due to the greater computational cost required in handling ambiguity (Fig. 1d). To complement this measure of neural entropy on the behavioural level, we use a novel implementation of entropy on individuals' RT distributions to quantify the irregularity in behavioural responses (Fitousi, 2023) (Fig. 1b), which we hypothesize to show the same pattern of results as SE.

Lastly, being that the unambiguous condition should elicit normal, automatic self-recognition responses (Volpara et al., 2022; Kotlewska et al., 2023), we predict that it is in the ambiguous condition specifically that individual responding will diversify, showing high intra- and inter-subject variability. In this condition, we should therefore see positive correlations between neural and behavioural measures of speed and entropy, representing different degrees of ambiguity processing capacity and speed across subjects (Fig. 1c).

2. Methods

2.1. Participants

The datasets used in this study consisted of behavioural and neural data from 27 participants, originally described in the 2023 paper by Wolman and colleagues (Wolman et al., 2023). Of these participants, two were excluded due to problems with task-state EEG recordings. Another participant was removed due to irregular responding in the discontinuous task, leaving 24 participants for the present analyses. All participants provided informed consent and the study approved by the local Research Ethics Board of the University of Ottawa Institute of Mental Health Research at The Royal (REB #2018,054). For additional detail on task design and methods beyond what is described here, see (Wolman et al., 2023).

2.2. Stimuli

The self-other facial morphing task uses images of participants' faces and with morphing software, combines them with that of a stranger matched by age, sex, and race (Keenan et al., 2000; Kircher et al., 2001; Tsakiris, 2008). Photos of participants with neutral facial expressions were collected prior to the study and were morphed with the face of a stranger matched by race and sex. For each participant, the two faces were morphed in 1% steps from 0% (other) to 100% (self), generating a

continuum of 101 pictures from the participant's face to the stranger's face. These images were used in two tasks, the discontinuous task and the continuous task.

2.3. Discontinuous morphing task

In the discontinuous task, morphed images in 10% morphing steps were selected from 0% to 100% self. Between 40% and 60%, images in 5% steps were selected to gather more evidence from images with greater self-ambiguity. These 13 images were presented 12 times for 200 ms in a random order for a total of 156 trials per participant. Following each image, participants were instructed to press "1" if the picture represented self (morphing steps >50%) or to press "2" if the picture represented other (morphing steps <50%). There was no time limit to provide a response. After the response was given, a fixation cross was presented for 800 ms preceding the next trial. Response time (RT) and accuracy were recorded.

In all analyses of data from the discontinuous task, the 50% morphing trials were excluded, due to the absence of a correct response. This was particularly important for the drift diffusion analysis which requires that one of the two choice alternatives to be correct. For consistency, this was carried over to the other RT analyses. The remaining data was separated into two conditions, ambiguous and unambiguous. The ambiguous condition consisted of 40%, 45%, 55%, and 60% morphing steps, while the unambiguous condition consisted of 0%, 10%, 20%, 30%, 70%, 80%, 90%, and 100% morphing steps. The main results are reported for these conditions, however an analysis of three morphing conditions, ambiguous, self (70%, 80%, 90%, and 100% morphing steps), and other (0%, 10%, 20%, and 30% morphing steps) was also performed, which is described in the supplement.

2.4. Continuous morphing task

In the continuous task, all 101 morphed images were presented sequentially from self to other or vice versa, each for a duration between 0.1 s to 0.15 s. These were presented to participants as video clips ranging from 10.1 s to 15.15 s in length. Between stimulus presentations, there was an intertrial interval jittered between 4 s and 6 s. Each participant completed a total of 200 trials, 50 morphing from self to other (SO) and 50 morphing from other to self (OS), in addition to two control conditions of 50 trials each morphed from an unknown face to a famous face (UF) and from a famous face to an unknown face (FU). Of these 200 trials, 180 trials were report trials and required participants to make a response, while the remaining 20 trials (five in each condition) involved simply watching the stimuli without responding. During the 180 report trials, participants were instructed to identify when they stopped seeing the first face by pressing a key, and to press a different key when they started seeing the second face. All 200 trials were recorded in EEG.

2.5. Behavioural analyses

2.5.1. Hierarchical drift diffusion modelling

As a two-alternative forced choice task, the discontinuous morphing task is ideally suited to computational modelling with the hierarchical drift diffusion model (HDDM) for a fine-grained analysis of the decision process in ambiguous vs. unambiguous morphing conditions. There are four principal parameters in the model, the drift rate (v) or the evidence accumulation process, the starting point (z) of the drift rate, the threshold/boundaries (a) of the diffusion, and the non-decision time (t) reflecting time allotted to non-decision processes such as sensory or motor functions. More complex models also include variability in the v , t , and z parameters.

In contrast to the classic DDM, HDDM allows estimation of both group-level and subject-level parameters, where the subject-level parameters are drawn from the group-level distributions allowing for a

more realistic and flexible model (Wiecki et al., 2013; Vandekerckhove et al., 2011). HDDM was performed in Python using the HDDM package, an open-source Python toolbox for conducting hierarchical Bayesian estimation of drift diffusion model parameters (Wiecki et al., 2013). The dockerHDDM environment was used on a MacBook M2 to solve compatibility issues (Pan et al., 2025). As per previous research (Wiecki et al., 2013; Vandekerckhove et al., 2011; Matzke and Wagenmakers, 2009), weakly informative priors (HDDM package default priors) were used for parameter estimation to constrain parameter values to psychologically plausible ranges (see Wiecki et al., 2013) for a detailed discussion of prior specification in HDDM).

We sought to contrast the evidence accumulation process during decision-making in the discontinuous task for the ambiguous vs. the unambiguous morphing conditions. Specifically, the rate of evidence accumulation, the drift rate (v), and the drift rate variability (sv) were estimated separately for the ambiguous and unambiguous conditions. The data were accuracy-coded, meaning the boundaries of the evidence accumulation process represented correct (upper boundary) or incorrect (lower boundary) responses.

2.5.2. Cumulative residual entropy of response times

The entropy of RT distributions can hold valuable information about various psychological characteristics of responses such as processing capacity or efficiency (Townsend and Ashby, 1978; Kahneman, 1973). For a thorough investigation of response behaviour under self-ambiguity, we sought to compute the entropy in participants' RTs in ambiguous and unambiguous conditions. The most widely used measures of entropy in information theory (e.g. Shannon's entropy) ran into issues with the computation of differential entropy for continuous distributions, such as those of RTs (Rao et al., 2004). Based on Shannon's entropy, Rao and colleagues (Rao et al., 2004) developed the measure of cumulative residual entropy (CRE) to solve these issues. CRE provides many advantages, mainly that it can be calculated on both continuous and discrete distributions, and that it can be used on empirical data (Fitousi, 2023). CRE was computed according to the method outlined by Fitousi (Fitousi, 2023) using the following formula, where $S(t)$ denotes the empirical survivor distribution of the RTs:

$$CRE = - \int S(t) \log_{10} S(t) dt$$

The integration was performed using the AUC function in the MESS package (Ekström, 2023) in R (R Core Team, 2025) according to Fitousi (Fitousi, 2023). This was done separately for the ambiguous and unambiguous morphing conditions of the discontinuous task to generate two measures of entropy per participant. Here, the survivor function, in the context of RT distributions, describes the probability that a response has not occurred. CRE then captures the uncertainty in whether a response has occurred over the response time. Although variance and entropy are related, variance captures dispersion around the mean, whereas CRE captures the temporal uncertainty. As shown by Fitousi (Fitousi, 2023), CRE increases when uncertainty about response completion persists over time, reflecting slower information gain and prolonged survival of the response process, and can dissociate from both mean RT and variance.

2.6. EEG processing and analysis

The EEG data was recorded and preprocessed for the 2023 study by Wolman and colleagues. Neural activity during the continuous task was recorded using Ag/AgCl electrodes through a 64-channel Brain Vision EasyCap (according to the International 10–20 System) referenced to the right mastoid. The data was sampled at 1000 Hz, downsampled to 500 Hz, then bandpassed from 1 to 50 Hz. All preprocessing prior to the analyses done in this paper was performed in the EEGLAB toolbox for MATLAB R2017b (Wolman et al., 2023). In addition to EEG recordings during the continuous task, resting-state activity was recorded for seven

minutes while participants were fixating a white cross on a gray background.

For the analysis of neural activity during ambiguous vs. unambiguous conditions, the EEG activity on each report trial (180 per participant) was divided into 3 s intervals: the first 3 s, the last 3 s, and the middle 3 s surrounding the 50 % morphed image. Thus, three separate timeseries were generated for each participant, one for self morphings (first 3 s in SO trials, last 3 s in OS trials), one for other morphings (last 3 s in SO trials, first 3 s in OS trials), and one for ambiguous morphings (middle 3 s) in order to represent all morphing categories. To contrast rest and task activity, a 3 s interval was extracted from the resting-state activity from 207 to 210 s of the seven-minute recording. These processing steps were done in the EEGLAB v2025 toolbox (Delorme and Makeig, 2004) for MATLAB R2024b (The MathWorks, 2024).

2.6.1. Peak frequency sliding

To investigate the dynamics of internally oriented cognitive processing in relation to discrimination of ambiguous stimuli, we focused our analysis on moment-to-moment theta band (5–8 Hz) frequency dynamics quantified via peak frequency sliding (PFS). PFS was performed on the three task-state timeseries as well as the resting state timeseries for each participant on all 64 channels, using a method developed by Cohen (Cohen, 2014), through the frequencySliding MATLAB function (Wolff A., n.d.). The EEG timeseries were first bandpass filtered using a finite impulse response (FIR) filter, with a 15 % transition width, after which the phase angle timeseries was extracted from the analytic signal, computed by Hilbert transform. To get the instantaneous frequencies, the first derivative of the phase angle timeseries was computed, and this timeseries was then smoothed with a median filter of order 20 in order to minimize noise while maintaining theta frequency dynamics. The resulting PFS timeseries provides a phase-derived instantaneous frequency estimate of the oscillation at each time point, capturing non-stationary theta-band frequency evolution over time.

The mean value of the PFS timeseries across all 64 channels was taken for each timeseries (ambiguous, self, other) for the 180 report trials, as well as for resting-state, to get four global theta PFS values per participant. To contrast the unambiguous conditions against the ambiguous, the PFS values for self and other were averaged. Note that averaging the instantaneous frequency (PFS) timeseries is not equivalent to estimating a single peak frequency from the window's power spectrum (as is often done for peak alpha frequency): the spectral peak is the frequency with maximal power aggregated across time, whereas the mean instantaneous frequency is a time-averaged summary of the frequency trajectory and can diverge under non-stationarity or transient oscillatory bursts (Cohen, 2014; Hua et al., 2022). Under stationarity and approximately constant amplitude, these measures would be expected to converge; however, theta-band activity during task performance is not assumed to be stationary (Vidaurre et al., 2023).

As control measures to assess specificity, we also computed alpha-band PFS (8–13 Hz) using the same procedure as theta PFS, and theta-band power from the analytic signal as the squared magnitude of the Hilbert-transformed theta-band time series and averaged across time and electrodes to match the aggregation used for PFS (Wolff A. n.d.).

2.6.2. Spectral entropy

Spectral entropy (SE) is a frequency domain measure of entropy representing the irregularity and disorder across the various frequencies of an EEG power spectrum (Faust and Bairy, 2012). Lower SE values correspond to more regular and structured neural activity, characterized by concentration of power in fewer frequency components. This has been shown to correlate with improved cognitive functioning and increased efficiency in task performance (Keshmiri, 2020). To this end, we used the spectralEntropy function in MATLAB, which computes the SE of a signal by taking the Shannon entropy of its normalized power distribution in the frequency domain (Faust and Bairy, 2012). SE was calculated for the bandpass frequency range (1–50 Hz) of the task and

rest timeseries, globally across all 64 channels, which were then averaged over trials to give one value per participant in each condition: ambiguous, unambiguous, and rest.

2.7. Statistical analysis

To test hypotheses on the HDDM parameters, the posterior distributions of the v and sv parameters were compared directly between ambiguous and unambiguous conditions using Bayesian inference (Wiecki et al., 2013; Makowski et al., 2019). Analogous to one-tailed frequentist hypothesis tests, comparing posterior distributions provides the posterior probability that two parameters are different from one another in the hypothesized direction (Makowski et al., 2019). Thus, posterior probability values (P) greater than or equal to 0.95 were deemed statistically significant, indicating that there is a probability ≥ 0.95 that the parameters differ in accordance with the hypothesis of interest. The difference between conditions was further confirmed by using the percentage of the 95 % highest density interval (HDI) in the region of practical equivalence (ROPE) method as described by Kruschke (Kruschke, 2015; Kruschke, 2018). The ROPE was defined between $[-0.2, 0.2]$ to assess non-zero differences between parameters (Pan et al., 2025).

All remaining statistical analyses and data processing were performed in R (R Core Team, 2025). To compare RT CRE, mean theta PFS, and SE between ambiguous and unambiguous conditions, paired two-sample hypothesis tests were used. For the purposes of comparing rest vs. task neurodynamic measures, theta PFS and SE values for the ambiguous and unambiguous conditions were averaged and compared to the resting-state values for the same measures. Shapiro-Wilk tests of normality were performed on the distributions of the differences between conditions; measures that were normally distributed were compared with paired-samples t -tests, and those with skewed distributions were compared with Wilcoxon tests. To assess the relationship between brain and behaviour, behavioural and EEG measures were further investigated using Pearson correlations and linear regressions.

3. Results

3.1. Behavioural data I: High ambiguity is linked to longer response times and lower accuracy

Response time and accuracy data for each subject were compared between ambiguous and unambiguous morphing conditions using paired samples t -tests. RTs were significantly longer in the ambiguous condition compared to the unambiguous condition, $t(23) = 6.89$, $p < .001$, $d = 1.41$, with significantly lower average accuracy $t(23) = -13.85$, $p < .001$, $d = -2.83$ (Table 1), highlighting the increased difficulty of the task when presented with ambiguous stimuli.

3.2. Behavioural data II: Evidence accumulation is impaired for ambiguous stimuli

To further analyze the processing disadvantage for the ambiguous facial stimuli, three hierarchical drift diffusion models were estimated to

Table 1

Mean response time and accuracy for ambiguous and unambiguous morphing conditions.

Condition	RT (s) ^{***}	Accuracy ^{***}
Ambiguous	0.62 [0.18]	0.67 [0.11]
Unambiguous	0.48 [0.095]	0.95 [0.029]

Note. Mean [standard deviation] RT and accuracy values per condition for $n = 24$ subjects. Condition means were compared using paired-samples t -tests, yielding longer RTs and lower accuracy for ambiguous vs. unambiguous stimuli.

^{***} $p < .001$.

determine which explained the behavioural data best. Table S1 shows a comparison of these models; all models estimated separate drift rates and drift rate variabilities for ambiguous and unambiguous conditions. Model AEsV was chosen as the best representative model due to its lower DIC and its ability to measure subject-level variability in drift rate (as opposed to group-only). For a thorough comparison of the three models, as well as an assessment of model convergence, see supplementary materials.

The chosen model estimated ambiguous vs. unambiguous drift rate (v_A , v_E), ambiguous vs. unambiguous drift rate variability (sv_A , sv_E), threshold (a), starting point (z), and non-decision time (t) per subject. The HDDMTruncated function was used rather than the standard HDDM function due to its ability to estimate subject-level variability parameters. For this model, 100,000 Markov Chain Monte Carlo (MCMC) samples with 25,000 burn-ins were simulated for four chains to ensure adequate convergence, given that individual variability parameters in HDDM are notoriously harder to converge (Ratcliff et al., 2016; Wiecki et al., 2013).

We predicted that the decreased accuracy and longer RTs in the ambiguous condition would reflect in the v and sv parameters of the HDDM with slower, more variable drift rates. In accordance with these hypotheses, drift rate was significantly lower in the ambiguous condition ($v_A = 1.77$) than the unambiguous condition ($v_E = 3.45$; Fig. 2a). For the given sample there is a posterior probability of $P_{A < E} = 1.0$ that drift rates are lower for ambiguous morphings than unambiguous morphings, and this difference is significant ($M_E - A = 1.7$, 95 % HDI [1.4, 2], [0 % in ROPE]; Figure S2a). Further, analogous to the drift rate, drift rate variability was higher in the ambiguous condition ($sv_A = 1.70$) than in the unambiguous condition ($sv_E = 0.83$), with a posterior probability of $P_{A > E} = 1.0$ (Fig. 2b). The difference between unambiguous and ambiguous sv in the given sample is significantly different from zero ($M_E - A = -0.87$, 95 % HDI [-1.3, -0.47], [0 % in ROPE]; Figure S2b). In sum, drift rates towards the correct boundary were much slower and more variable across trials in the ambiguous condition compared to the unambiguous condition, demonstrating that greater ambiguity impairs processing efficiency in the identification of facial stimuli.

3.3. Behavioural data III: Cumulative residual entropy (CRE) of RTs is increased in response to ambiguity

Given the increased drift rate variability during the ambiguous condition, could this also imply that subjects' RTs in this condition have a higher degree of randomness? To test this assumption, we calculated the entropy (CRE) of the RT distributions of each participant for both conditions to measure of the randomness and disorder in response probabilities. We found that participants' RTs had greater entropy for ambiguous morphings ($Mdn_A = 0.36$) than unambiguous morphings ($Mdn_E = 0.22$) with a strong significant difference between conditions, $W = 291$, $p < .001$, and large Wilcoxon effect size, $r = 0.82$ (Fig. 2c). The increased RT CRE in processing ambiguous stimuli implies that there is a greater computational cost than for unambiguous morphing steps, similar to the results for v and sv from our drift diffusion model.

3.4. Behavioural data IV: Dissociation of entropy and drift rate under ambiguous conditions

Three measures were computed to characterize response behaviour in ambiguous vs. unambiguous conditions: v , sv , and CRE. RT mean and CRE are known to be independent features of RT distributions (Fitoussi, 2023), but does that relationship hold between drift parameters and CRE as well? To further examine this, the drift parameters were each correlated with the CRE, separately for each condition. In the ambiguous condition, there was no significant correlation between drift rate and entropy, $r(22) = -0.27$, $p = .27$, nor was there a significant correlation between drift rate variability and entropy, $r(22) = -0.088$, $p = .68$. In

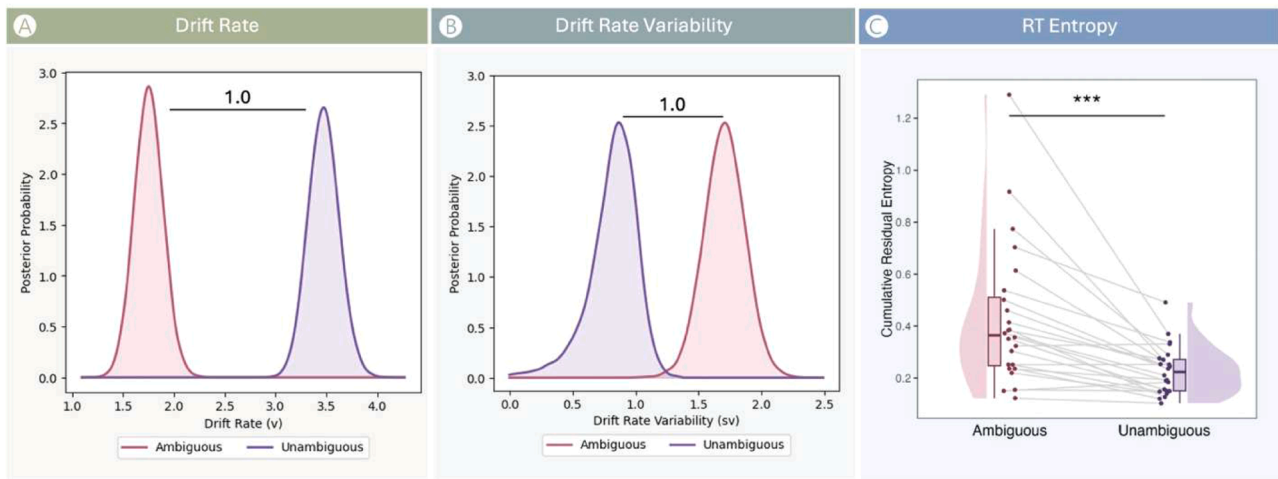


Fig. 2. Differences in Ambiguous and Unambiguous Morphings for Behavioural Measures. Comparisons of behavioural measures between ambiguous (pink) and unambiguous (purple) conditions from the discontinuous task. Posterior distributions of ambiguous vs. unambiguous drift rate (v) (A) and drift rate variability (sv) (B). Posterior probabilities of ambiguous $v <$ unambiguous v and ambiguous $sv >$ unambiguous sv are highly significant ($P_s = 1.0$). C) Raincloud plots depicting cumulative residual entropy (CRE) values for participants' RT distributions in ambiguous vs. unambiguous conditions. Points represent subject-level values; values for the same participants are connected with grey lines. The difference between conditions was evaluated using a paired Wilcoxon Test; ambiguous RT CRE was significantly greater than unambiguous RT CRE. *** $p < .001$.

the unambiguous condition, CRE's correlation with drift rate remained nonsignificant, $r(22) = -0.42$, $p = .08$. In contrast, drift rate variability showed a positive correlation with entropy, $r(22) = 0.68$, $p < .001$. The dissociation in the relationship between RT entropy and drift rate variability across the two conditions highlights fundamental cognitive processing differences in decision-making between ambiguous and unambiguous morphed images. This may reflect that entropy is no longer related to the evidence accumulation process when stimuli are more ambiguous.

3.5. Neurodynamic data I: Theta peak frequency decreases while sample entropy increases from rest to task

To examine the effect of task on EEG activity, resting-state activity and continuous task activity were compared for the measures of interest, theta PFS and SE. There was a significant decrease in theta PFS during the continuous task ($M_T = 5.39$, $SD_T = 0.26$) compared to rest ($M_R = 6.54$, $SD_R = 0.31$) with a large effect size, $t(23) = 19.74$, $p < .001$, $d = 4.03$ (Fig. 3a). Conversely, SE significantly increased from rest ($M_R = 0.61$, $SD_R = 0.063$) to task ($M_T = 0.64$, $SD_T = 0.049$) with a moderate effect size, $t(23) = -3.24$, $p = .004$, $d = -0.66$ (Fig. 3b). The decrease in theta PFS signals a shift towards slower frequencies with longer phase cycles, while the increase in entropy (SE) suggests decreased

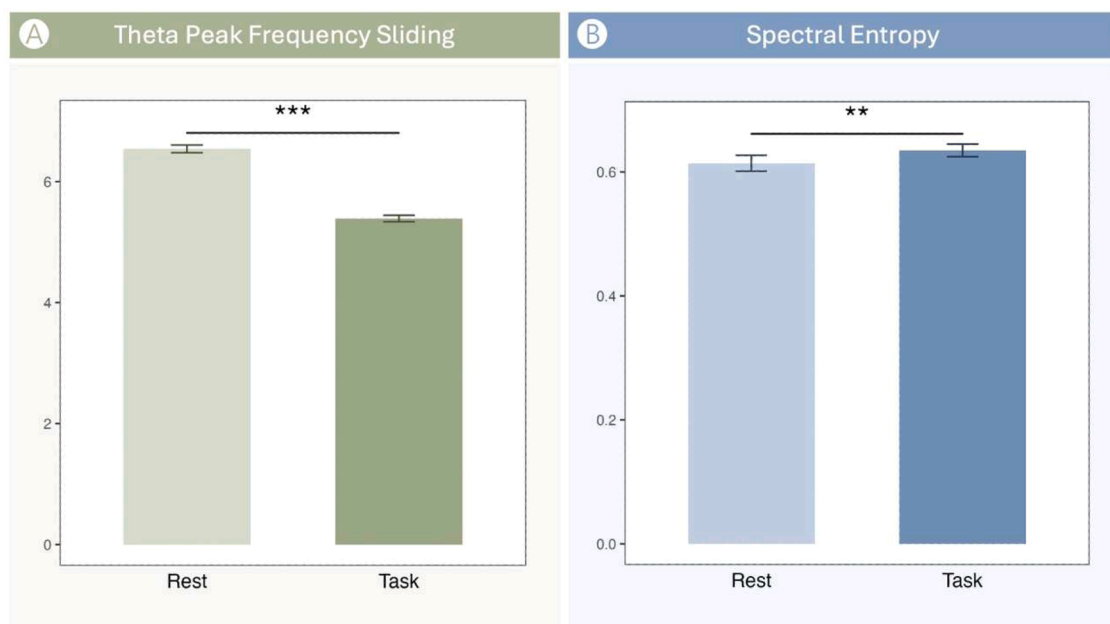


Fig. 3. Differences in Rest and Task Neurodynamic Measures. Continuous task neurodynamic EEG measures of theta peak frequency sliding (PFS) and spectral entropy (SE). Bars show a significant decrease in theta PFS (A) and increase in SE (B) from rest to task. The rest–task differences were assessed using paired-samples t -tests. ** $p < .01$. *** $p < .001$.

predictability of neural activity during the task state compared to the resting state.

3.6. Neurodynamic data II: Ambiguous stimuli elicit lower theta PFS and higher entropy in neural activity

During the task, the ambiguous condition was predicted to elicit lower PFS theta and greater spectral entropy than the unambiguous condition. Indeed, theta PFS was decreased for ambiguous morphings ($M_A = 5.37$, $SD_A = 0.25$) than unambiguous morphings ($M_E = 5.40$, $SD_E = 0.27$), with a significant large effect, $t(23) = -4.06$, $p < .001$, $d = -0.83$ (Fig. 4a). In contrast, alpha PFS did not differ between ambiguous and unambiguous conditions ($p > .05$). A lower theta PFS suggests poorer internal information uptake in the ambiguous condition than in the unambiguous condition, pointing to inferior precision for stimulus processing compared to unambiguous stimuli. Scalp topographies of theta PFS are provided for visualization in Figure S6.

Moreover, as expected, SE values were significantly higher in the ambiguous condition ($M_A = 0.64$, $SD_A = 0.048$) than in the unambiguous condition ($M_E = 0.63$, $SD_E = 0.050$), $t(23) = 6.07$, $p < .001$, with a large effect size, $d = 1.24$ (Fig. 4b). Participants had greater randomness and less predictability or higher uncertainty in their neural activity when viewing ambiguous stimuli, implying processing difficulties in the identification of these faces when compared to the unambiguous stimuli.

3.7. Relating behavioural and neurodynamic data

Both behavioural and neurodynamic data demonstrate the differential impact of high vs. low ambiguity in processing morphed facial stimuli. Thus, we sought to relate these measures within the two conditions. For measures indexing behavioural and neural speed, drift rate values were analyzed with task, rest, and rest-task difference theta PFS values. For measures indexing behavioural and neural entropy, RT CRE was analyzed with task, rest, and rest-task difference SE values. The linear relationships between these measures were first established using Pearson correlations. All correlations with unambiguous drift rates and RT CRE were nonsignificant ($ps > 0.05$). Additionally, behavioural measures (drift rate, RT CRE) were not significantly related to the

control measures (alpha PFS and theta power) across task, rest, or rest-task difference values ($ps > 0.05$). Task-state and rest-task theta PFS did not show a significant relationship with drift rate in either condition, nor did resting- or task-state SE alone significantly correlate with RT CRE, indicating that the observed effects were specific to resting-state theta dynamics and to rest-task modulation of SE. In the ambiguous condition, drift rates were significantly correlated to rest theta PFS ($r(22) = 0.45$, $p = .028$) and RT CRE was significantly correlated to rest-task difference in SE ($r(22) = 0.60$, $p = .002$)—these relationships are further explained below.

A linear regression was run to test whether resting-state theta PFS is associated with participants' ambiguous drift rates, based on their significant correlation. A trend analysis further confirmed that there was a significant linear relationship between rest theta PFS and ambiguous drift rate. The fitted equation was: $y_v = 0.48x_{PFS} - 1.37$. Fig. 5a shows resting-state PFS theta activity had a positive linear relationship with drift rates in the ambiguous condition, $F(1, 22) = 5.52$, $p = .028$, accounting for $R^2 = 20.1\%$ of the variation in drift rates with adjusted $R^2 = 16.44\%$, a moderate effect size according to Cohen (Cohen, 1988). A one-point increase in a participants' resting theta PFS was associated with a 0.48-point ($SE = 0.20$) increase in ambiguous drift rate.

In addition, there was a significant correlation between the rest-task difference in SE and CRE of ambiguous condition RTs. This was further investigated with a linear regression to test if neural entropy was a significant regressor of behavioural entropy in the ambiguous condition, with the equation $y_{CRE} = 4.91x_{SE} + 0.32$. The linear relationship between the rest-task difference in SE and ambiguous RT CRE was significant, $F(1, 22) = 12.42$, $p = .002$, with a large effect size, accounting for $R^2 = 36.1\%$ of the variation in CRE with adjusted $R^2 = 33.2\%$ (Fig. 5b). A one-point increase in a participants' rest-task difference in SE was related to a 4.91-point ($SE = 1.39$) increase in RT entropy for ambiguous facial stimuli.

In summary, our results demonstrate significant positive relationships between neural and behavioural measures for ambiguous facial morphings. We show that faster theta PFS is related to faster information uptake (drift rate), while higher neural entropy is related to higher values of entropy in response times. These findings are supported by exploratory control analyses reported in the Supplementary Materials.

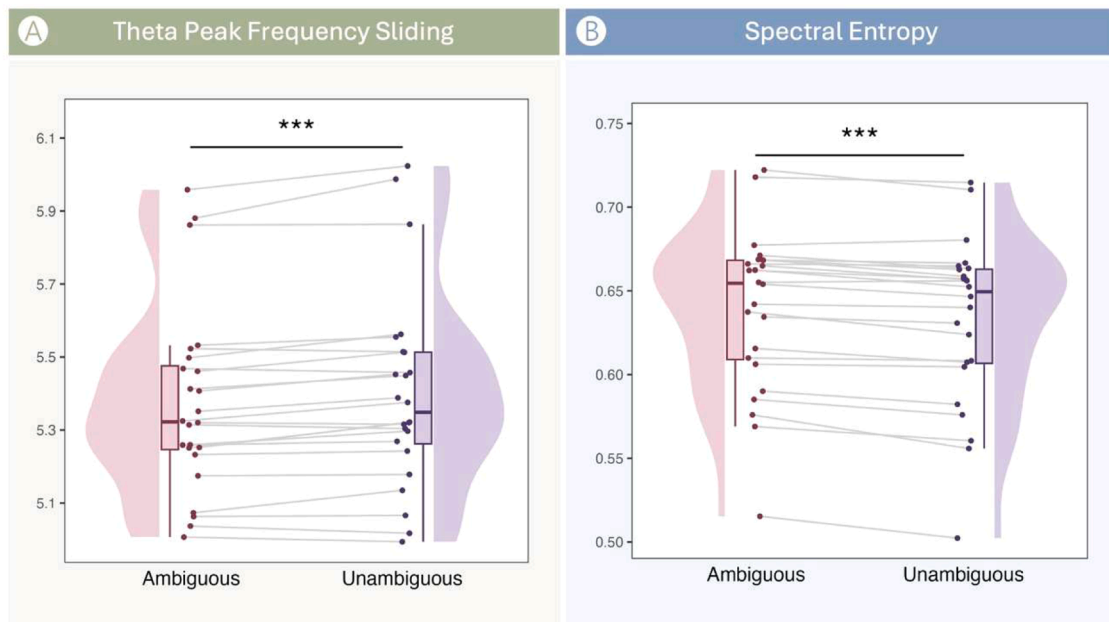


Fig. 4. Differences in Ambiguous and Unambiguous Morphings for Neurodynamic Measures. Raincloud plots depicting ambiguous vs. unambiguous values for theta PFS (A) and SE (B) from the continuous task. Points represent subject-level values; values for the same participants are connected with grey lines. The differences between conditions were evaluated using paired-samples t -tests, where the ambiguous condition elicited significantly slower theta PFS (A) and greater SE (B). *** $p < .001$.

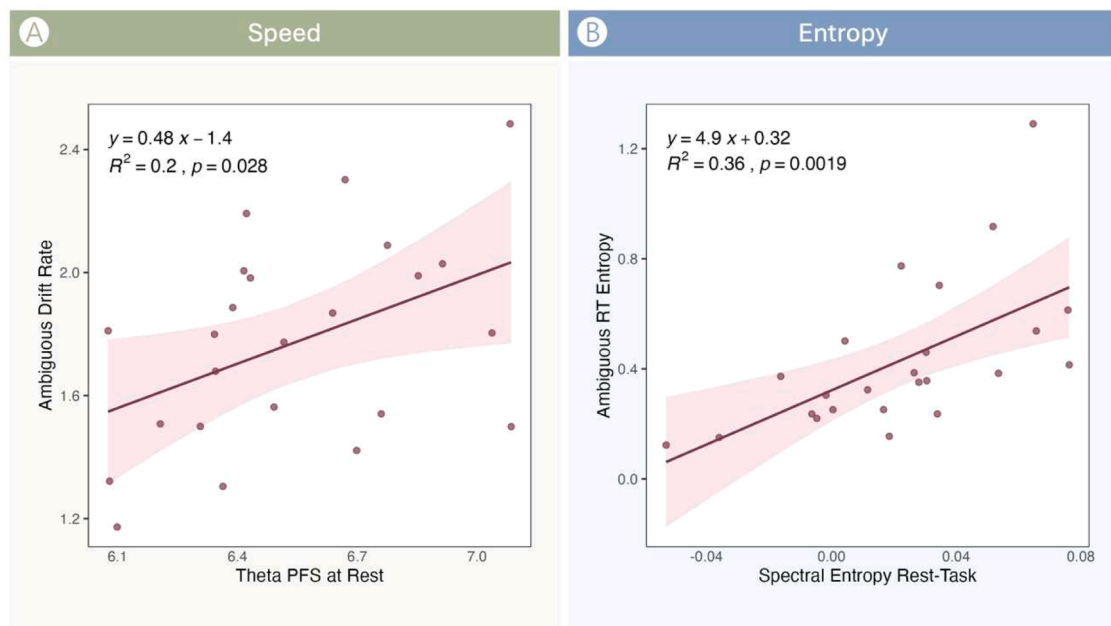


Fig. 5. Linear Models of Neurobehavioural Relationships. Linear regressions illustrating the relationship between behaviour in the ambiguous condition of the discontinuous task and EEG activity for speed and entropy measures. Scatterplots are shown with lines of best fit and 95% confidence intervals. A) The strength of resting-state theta peak frequency has a significant positive linear relationship with ambiguous drift rate. B) Neural entropy is significantly associated with behavioural entropy such that the greater the increase in participants' spectral entropy from rest to task, the greater their ambiguous RT cumulative residual entropy in the discontinuous task.

Together, these results suggest a strong neurodynamic and temporal shaping (theta PFS and SE) of the behavioral measures (drift rate and RT entropy) during cognitive decision-making under self-other ambiguity.

4. Discussion

We conducted a neurobehavioural analysis of the dynamics of decision-making of ambiguous and unambiguous self-face stimuli to determine how perceptual ambiguity affects self-face recognition. Our findings can be classified into two main dynamic categories, speed and entropy. Firstly, we sought to measure neural and behavioural speed of information processing. Our results confirmed our hypotheses both neurally and behaviourally. We found a slower stimulus-related processing speed for ambiguous stimuli, as evidenced by slower drift rates and theta PFS. These measures were linearly related: ambiguous drift rate was modulated by resting-state theta PFS such that greater theta PFS was associated with better evidence accumulation. Secondly, we analyzed ambiguity processing using entropy as a measure of both the unpredictability and processing capacity of responses and the spectral distribution of neural activity (spectral entropy/SE). Confirming our predictions, RT entropy and SE were both significantly increased in the ambiguous condition, with more positive rest-task SE differences strongly relating to higher entropy of responses. Together, these findings make a direct connection between behavioural and neural measures of ambiguous decision making through their shared dynamic features, e.g., entropy and speed.

4.1. Evidence accumulation from internal sources is key for the disambiguation of perceptual stimuli

Drift diffusion modelling (DDM) has consistently shown that more complex, uncertain, or difficult decisions have slower drift rates towards the correct decision boundary and that these differences are reflected in the brain (Sun et al., 2025; Ratcliff et al., 2016; Harris and Hutcherson, 2022; Voss et al., 2013). In line with previous data, our results show that introducing stimulus ambiguity decreases drift rates and accuracy while

increasing RTs, implying a deficit in cognitive processing compared to unambiguous stimuli. It is worth noting that much of the research on ambiguity in decision-making has focused on either probability ambiguity in economic contexts (Wu et al., 2021), or ambiguous sensory stimuli (Kelly and O'Connell, 2013). Of interest here is perceptual ambiguity, which recruits fundamentally different cognitive processes since it relies on an individual's perceptual representations of the content of the stimuli (Gao et al., 2025; Mercier and Cappe, 2020). As a result, disambiguating such higher-level perceptual stimuli is inherently more subjective and internally-driven than for lower-level perceptual detection (e.g., luminance/contrast detection), and involves different considerations than economic ambiguity (Gao et al., 2025; Van Ede and Nobre, 2024). Previous research by Sun et al. (Sun et al., 2025) on perceptual ambiguity has focused on ambiguity in facial expressions and the emotional content of such stimuli, demonstrating slower evidence accumulation for the classification of faces with high emotional ambiguity. Our results add to this research, demonstrating that even for a highly salient, overlearned stimulus such as self-face, increasing ambiguity interferes with normal self-face recognition by slowing down evidence accumulation and increasing variability in drift rates.

How can we explain the perceptual difficulties observed in discriminating ambiguous morphings of the self-face with an unknown other? A theory of face recognition originally proposed by Valentine (Valentine, 1991) posits that all the faces one has encountered are stored as perceptual representations in a multidimensional "face space". It follows that the self-face lies somewhere within the face space in memory as well. Tying this to perceptual ambiguity processing (Klink et al., 2012), to discriminate ambiguous self-face stimuli, perception of the morphed stimulus is constantly fluctuating between self and other. We postulate that every time the facial representations are accessed from the face space and evaluated within working memory (Van Ede and Nobre, 2024; Nobre et al., 2004), evidence in favour of one face or the other accumulates and a decision is made. Such top-down projections of internal information have been demonstrated neurally in response to perceptually difficult facial stimuli (Karimi-Rouzabehani et al., 2021; Fan et al., 2020).

In a recent paper, van Ede and Nobre (Van Ede and Nobre, 2024) have demonstrated this internal drift diffusion process, and here we provide further support for this perceptual mechanism in the context of self-other ambiguity processing. We postulate that each time a face drawn from the “face space” is considered against the morphed stimulus, it provides evidence in favour of the correct interpretation of the stimulus, in line with the fundamental logic behind the drift diffusion process (Ratcliff, 1978; Ratcliff and McKoon, 2008). For unambiguous morphings, the fluctuation of the perceptual representations of the faces terminates faster, reflected by faster drift rates. In contrast, for ambiguous morphings, the evaluation of the different facial interpretations is slower and more difficult since the ambiguous stimulus has less similarity to either face within the “face space”, providing weaker evidence for the decision.

4.2. Neurocomputational mechanisms of perceptual processing I: Ambiguity of morphed self-other stimuli slows information uptake

Using peak frequency sliding (PFS) in the theta band, we demonstrate the neurocomputational cost of the internal evidence accumulation process under ambiguous vs. unambiguous conditions. To reiterate, the mean PFS provides a measure of the strength of neural inputs (Cohen, 2014), which can be considered an index of the rate of neural information uptake/processing. In this way, PFS is said to act as a gain-control mechanism (Cohen, 2014), allowing the brain to adapt to situational demands. Faster PFS allows processing of faster and stronger inputs, allowing for increased information uptake, while slower PFS facilitates responding to slower and weaker inputs, decreasing quality of evidence and information uptake (Mierau et al., 2017; Cohen, 2014). For the case of theta PFS, this input comes from internal sources, such as internal task-unrelated thoughts (Hua et al., 2022; Wolff et al., 2019), which, in the present context is thought to be the internal representations of the facial stimuli as they are evaluated for the judgement each morphed image (explained above). In the ambiguous condition, there is a need for more evidence and more time to reach a decision, which manifests in the slowing down of both drift rates and theta PFS. Both measures demonstrate slower information uptake in this condition due to the weaker input provided by the uncertain stimuli, requiring longer processing to extract the information needed to make a proper decision. In the unambiguous condition, faster theta PFS and drift rates reflect the stronger evidence accumulation process with faster information uptake, consistent with more accurate responding.

To assess the specificity of theta PFS effects, we additionally computed alpha PFS (TH Grandy et al., 2013; Mierau et al., 2017), which did not differ between ambiguous and unambiguous conditions, highlighting the importance of internal input processing (as related to theta, while alpha is related to external input processing (Klimesch, 2012)) for the successful completion of the morphing task. For the first time, we show that gating of internal input, as measured by PFS in the theta band, may be a way for the brain to disambiguate perceptual stimuli. The importance of internal inputs in ambiguous decision making is further supported by other findings showing analogous recruitment of internal neural activity, e.g., spontaneous or resting state activity, when making decisions in situations without any clear external inputs serving as cues or guidance (Nakao et al., 2012; Nakao et al., 2013; Nakao et al., 2019).

Importantly, we further demonstrate a direct link between these theta dynamics and behavioural dynamic measures. There was a significant positive linear relationship between resting-state theta PFS and ambiguous drift rates, such that those participants with faster PFS tended to have better evidence accumulation (drift rates) in the ambiguous condition, and these effects were not replicated by alpha-band PFS. Resting-state neurodynamics have been shown to influence the dynamics of task-evoked activity and subsequent behavioural performance (Wolff et al., 2019; Nakao et al., 2013; Nakao et al., 2019; Huang et al., 2017; Northoff et al., 2010), particularly in the theta band (Wolff et al.,

2019). Moreover, these dynamics are thought to be the “common currency” of neural and mental states, as an integrated brain-mind framework (Northoff et al., 2025; Northoff et al., 2020). We lend support to this integrated brain-mind framework, showing that resting-state theta peak frequency dynamics influence behavioural task performance. Thus, we speculate that those with faster resting-state theta PFS have stronger and faster spontaneous internal input processing and are more cognizant of their neuro-mental state. This in turn seems to influence how efficiently they access and evaluate perceptual representations of their own and unfamiliar faces to disambiguate task-relevant stimuli. Therefore, we argue that the speed of information uptake provides a “common currency” (Northoff et al., 2025; Northoff et al., 2020) of neural and mental features (Fig. 6).

4.3. Neurocomputational mechanisms of perceptual processing II: Entropy as a measurement of capacity increases under self-other ambiguity

We determined, using DDM and PFS, that to solve the ambiguity of morphed faces, there is a higher demand for assessing and cycling through the perceptual representations of the two identities involved. Consequently, this process would require an increased information processing capacity, or in other words, increased Shannon’s entropy (Keshmiri, 2020; Shannon, 1948). By measuring the certainty of prediction of a system from the probability distributions of its activity, Shannon’s entropy quantifies the number of possible states/outcomes of the system (Keshmiri, 2020; Shannon, 1948; Rao et al., 2004). Systems with higher entropy would therefore allow for greater processing capacity to maintain a greater number of potential states in comparison to systems with lower entropy. In our analyses we applied this measure behaviourally using cumulative residual entropy (CRE) of RTs (Fitoussi, 2023; Rao et al., 2004), and neurally using spectral entropy (SE) (Keshmiri, 2020; Bachiller et al., 2014; Thul et al., 2016). Following our predictions, we found greater entropy in the ambiguous condition both in participants’ RTs and their spectral densities, demonstrating the increased processing demand for ambiguous stimuli. This is contrasted by the lower entropy in the unambiguous condition, in which the outcomes of each trial are more certain, relating to more regular and consistent spectral activity and response times.

This is the first paper to date to examine both behavioural and neural measures of Shannon’s entropy, and from these results, we propose that entropy acts as a “common currency” (Northoff et al., 2025; Northoff et al., 2020) in neural and mental perceptual processing for the disambiguation of stimuli. Consequently, we found that a stronger increase in SE from rest to task correlated significantly to higher RT entropy in the ambiguous condition, while those with smaller differences (or even decreases) in rest–task SE had lower RT entropy. Modulation of SE has been shown to be a marker of one’s ability to harness their mental resources to more efficiently tackle task-related challenges, where larger decreases in SE from pre- to post-stimulus were related to better task performance (Keshmiri, 2020; Tian et al., 2017). Our results are in line with these findings: constraining SE from rest to task helps to reduce entropy of responses. Hence, by adapting information capacity, rest–task SE modulation acts as a mechanism for more efficient processing of perceptual ambiguity as evidenced by CRE of RTs.

Although the “common currency” patterns for speed of information uptake and entropy might be construed as reflecting a trade-off between the two, we did not observe strong evidence for systematic trade-offs across individuals (Figure S5), nor for interactive effects between theta PFS and spectral entropy ($\Delta SE \times PFS$) in relation to behavioural measures (Table S2). Notably, exploratory correlational analyses (see Supplementary Materials) revealed that resting-state theta PFS was positively, rather than negatively, related to rest–task changes in spectral entropy (Figure S5), arguing against a simple trade-off between speed- and entropy-related neural dynamics. This positive association is not unexpected, given that neural dynamic measures such as theta PFS and spectral entropy are derived from the same underlying signal and

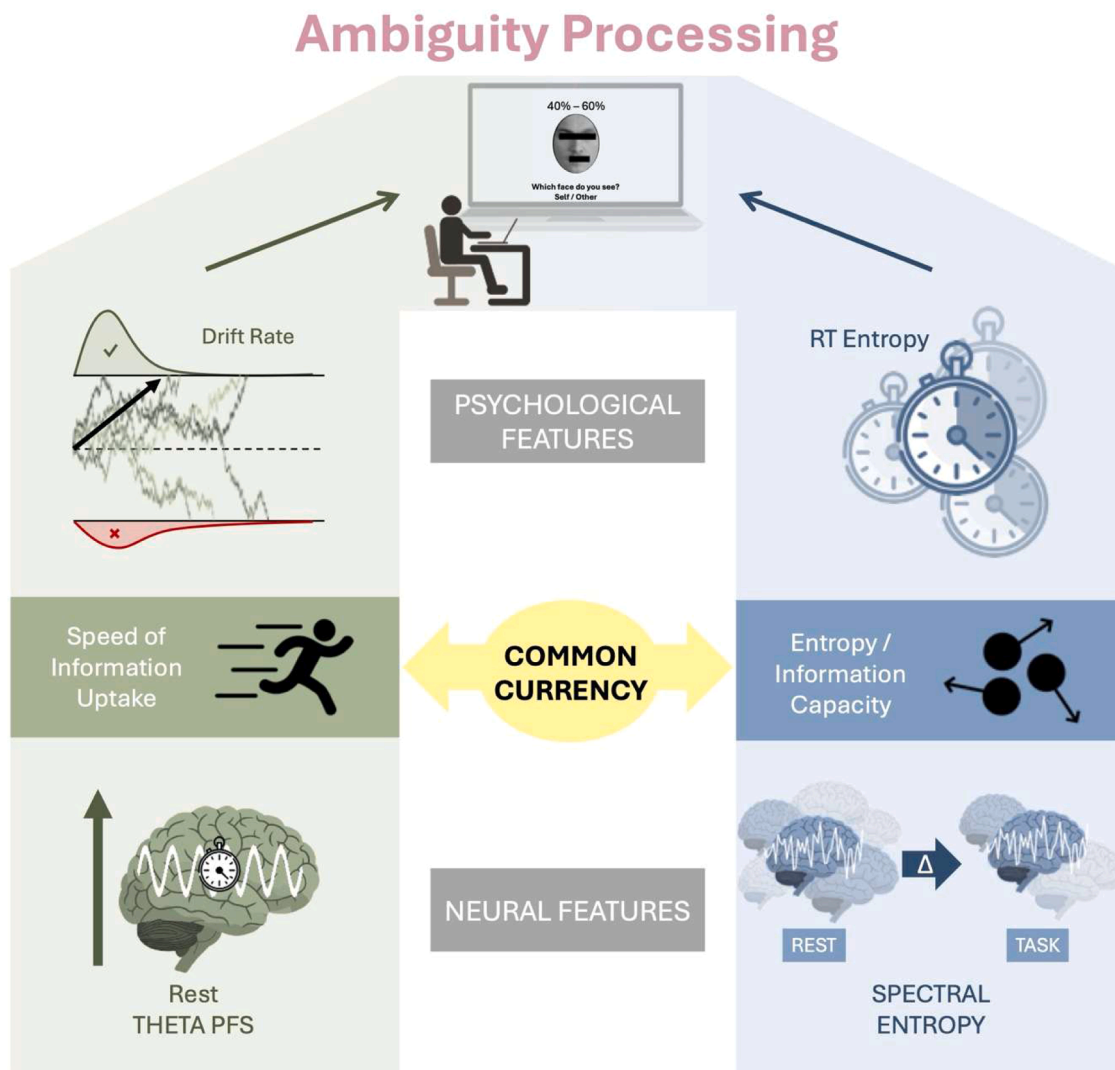


Fig. 6. *Neurocomputational Mechanisms for Perceptual Ambiguity Processing.* Speed of information uptake and capacity for information processing form the common currency between brain and behaviour for perceptual ambiguity processing. For ambiguous stimuli, evidence accumulation (drift rate) is influenced by intrinsic internal input gating (resting-state theta PFS), while SE modulation from rest to task shapes the predictability and order of response times. These results highlight the importance of intrinsic neurodynamic features for the efficient resolution of perceptual challenges.

thus share variance. Consistent with this, exploratory joint models and interaction terms ($\Delta SE \times PFS$) did not significantly vary with behavioural measures, providing no evidence that the relationship between one neural measure and behaviour depends on the level of the other (Table S2). Moreover, controlling for theta-band power (analytic amplitude squared) did not remove the primary neurobehavioural associations, suggesting that these effects are not readily explained by oscillatory amplitude alone. Analogous to DDM frameworks in which parameters such as drift rate and threshold are treated as largely independent (Ratcliff et al., 2016; Ratcliff and McKoon, 2008), we interpret theta PFS and entropy as indexing complementary but partially independent aspects of neural computation.

4.4. Differences in stimulus processing in ambiguous vs. unambiguous conditions

It is important to highlight that these two neurobehavioural relationships were found solely for ambiguous facial morphings, where the brain can no longer rely on automatic, overlearned self-recognition methods (Volpara et al., 2022; Kotlewaska et al., 2023; Miyakoshi et al., 2010; Keyes et al., 2010). To elaborate, the advantage of the

self-other facial morphing task is that it provides a way to explicitly consider the self-recognition process. Kotlewaska and colleagues (Kotlewaska et al., 2023) illustrated the automaticity of self-face recognition, finding an early reduction in theta activity in response to self-face perception. Here, by introducing ambiguity, self-recognition is no longer automatic, and facial recognition processes were more deliberate. This may explain why resting-state dynamics had a significant influence on *ambiguous* rather than unambiguous stimulus processing. The internal inputs from the brain's spontaneous or resting state activity are required to solve (or resolve) the ambiguity in the decision making of perceptually ambiguous stimuli neither conforming to either self or other faces. The balance of internal vs external inputs thus tilts towards internal inputs with decision making being more internally- than externally-guided (Wolff et al., 2019; Nakao et al., 2012; Nakao et al., 2013; Nakao et al., 2019).

The difference in processing between conditions is further supported by the non-significant correlations between RT entropy and drift rate variability in the ambiguous condition, in comparison to the unambiguous condition where entropy had a positive correlation with drift rate variability. We speculate that this dissociation within the ambiguous condition reflects a divergence from the predictable computational

patterns of the unambiguous condition where there is familiar, fast self-recognition. However, the relationships between entropy, drift rate, and variability need to be investigated more rigorously in future studies.

It is worth noting that the present task leverages self-other stimuli as a highly salient, overlearned stimulus class to facilitate analysis of the effects of increasing perceptual ambiguity, which is an inherent manipulation of task difficulty. Future work employing non-self control stimuli matched for ambiguity and difficulty will be required to determine the extent to which these effects are specific to self-related processing, rather than reflecting general mechanisms of ambiguity resolution. Thus, while our findings are interpreted in the context of self-other perception, they may also reflect neural mechanisms supporting the resolution of ambiguity more broadly.

4.5. Limitations & future directions

An important limitation of our study was that the behavioural analysis was performed on data from the discontinuous version of the self-other facial morphing task, while EEG recordings were performed during the presentation of the continuous version of the task. While both versions of the task are ultimately operationalizing the same cognitive process (self-face recognition) (Wolman et al., 2023), in the continuous task participants have prior knowledge of the nature of the next image and are aware that there are ambiguous stimuli in the middle of the video. As the video progresses, the allocation of attention to the images may change, biasing responses. Regardless, there were robust neural differences in ambiguous vs. unambiguous processing in both versions of the task, although it is possible that EEG and behavioural results would have been more consistent with each other had all the data been from the discontinuous task. This may explain why we only found neuro-behavioural relationships with resting-state activity. Future investigations using this task should consider this in their study design. Should they wish to use DDM with the discontinuous task, recording this task in EEG would allow the effects of both pre- and post-stimulus neurodynamics on the DDM parameters to be measured.

Further, in the present investigation, we focused on sensor-level, global measures of theta PFS to capture large-scale neural dynamics relevant to ambiguity processing. Scalp topographies of theta PFS (Figure S6) revealed broad, uniformly distributed patterns with similar spatial profiles across conditions, supporting the use of global rather than region-specific interpretations. While future work combining higher-density recordings and individual anatomical models could further examine spatial specificity, source localisation was beyond the scope of the current dataset.

5. Conclusion

We investigated the dynamics of self-face recognition under different degrees of perceptual ambiguity to determine the neural and behavioural processes involved in self-other discrimination. We demonstrate consistent evidence of a deficit under perceptually ambiguous conditions both in terms of speed and capacity/entropy of information processing. Our results help unveil some of the dynamic and computational solutions involved in the disambiguation of facial stimuli not clearly depicting the self or another's face. Thus, we propose a dynamic framework of the neurocomputational mechanisms for the resolution of the perceptual ambiguity of faces, whereby this decision process is dependent on dynamics of evaluating inputs from internal perceptual representations of facial identities. Importantly, the two key dynamic features, speed and capacity/entropy, provide a direct connection of the neural and behavioural levels during the information processing of perceptual self-other ambiguity.

Data availability

The EEG and behavioural dataset analyzed in this study was

originally published in Wolman et al. (2023) and is available upon request from the original authors. The present study reports secondary analyses based on those data; derived data produced for this manuscript are available upon reasonable request from the corresponding author.

Code availability

No new code, algorithms, or computational models were developed. All relevant details regarding data sources and analysis procedures are described in the Methods section of the manuscript. Custom scripts implementing standard analyses are available upon reasonable request from the corresponding author.

CRedit authorship contribution statement

Andrea Avramescu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Angelika Wolman:** Writing – review & editing, Resources, Data curation. **Georg Northoff:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the European Union's Horizon 2020 Framework Programme for Research and Innovation under the Specific Grant Agreement No 785907 (Human Brain Project SGA2), the EJLB–Michael Smith Foundation, the Canadian Institutes of Health Research (CIHR), the Start-up Research Grant at Hangzhou Normal University, and the NFRF/OBMRI Team Grant to G.N. The funding agencies had no role in the writing of the manuscript or in the decision to submit it for publication.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.neuroimage.2026.121839](https://doi.org/10.1016/j.neuroimage.2026.121839).

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